

COURSE 3 - DATA ENGINEERING 2026

C3-104A - EnergyOps Architecture Defense Gate

Protected independent C3-09 phase assessment | fresh energy readings domain.

Artifact	C3-104A
Status	2026 - LEARNER EDITION
Phase	C3-09 Production Capstone & Engineering Transition 15 lessons 36-48 focused hours
Prerequisite	C3-08 Gate PASS; all earlier C3 evidence retained.
Assessment boundary	C3-104 is the phase Gate only. C3-113 remains the later course-wide Final Production Gate.
Brand	Charcoal #1F2937 Teal #14B8A6 Soft Blue #3B82F6 AMOLED #000

House teaching rule: requirement -> mental picture -> small proof -> guided integrated build -> expected result -> why it worked -> break/fix -> changed independent case -> validation -> production reasoning -> evidence. Integration is not mastery until the learner can explain and recover the system independently.

Mission

EnergyOps needs a trustworthy energy readings data product with correction/new-record semantics, governed serving, production controls, controlled incident recovery and an independently defensible deployment path.

Required deliverables

- ADR with users/data product/measurable NFRs and one rejected architecture option.
- Source contracts with stable key/change semantics and reproducible synthetic input story.
- Batch + incremental implementation/design proving corrections/new records converge exactly once under replay.
- Bronze/Silver/serving contracts and justified Lakehouse/Warehouse boundary.
- Testable Spark component or executed distributed component with 10x-scale role and parity invariant.
- Orchestration with retry classes, partial rerun and one bounded historical backfill.
- Blocking quality controls plus independent key/count/value/business-total reconciliation.
- SLOs/logs/metrics/alerts and incident runbook.
- Least-privilege access, secret/federated-identity boundary and privacy-safe evidence.
- CI/CD with protected production change, rollback/roll-forward decision and post-deploy validation.
- Injected corruption/failure response: contain, diagnose, repair, replay/backfill and reconcile.
- Streaming include/exclude decision tied to latency requirement.
- Reproducible README/handoff and architecture defense.

Pass contract

PASS requires every required competency to PASS and zero unresolved critical failures. Every competency is judged independently; one strength cannot substitute for a missing required competency.

Required competencies

- A - requirements & architecture
- B - source/incremental data integrity
- C - transformation/serving correctness
- D - orchestration & recovery
- E - Spark/distributed parity
- F - quality & reconciliation
- G - observability & incident response
- H - security & governance
- I - CI/CD & change safety
- J - maintainability, reproducibility & architecture defense

Critical failures

- Fabricated live runtime/cloud/security evidence
- Secret/credential/PII leakage
- Wrong grain or duplicate/regressed current state under replay
- Unsafe or unbounded recovery that risks unrelated data
- Critical quality failure published
- Security claim tested only with over-privileged identity
- Deploy bypasses failed tests/required approval
- Incident closed without post-recovery reconciliation
- Protected review opened during active attempt

Boundary

This is C3-09 phase Gate C3-104. A pass does not equal Course 3 graduation; C3-113 remains the separate final production assessment.